

Erdős Problem 700(ii): An Unconditional Infinite Family

Nanocodex Erdős 700 project

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Abstract

For

$$f(n) = \min_{2 \leq k \leq \lfloor n/2 \rfloor} \gcd\left(n, \binom{n}{k}\right),$$

we prove that infinitely many composite integers n satisfy $f(n)^2 > n$. The proof has two parts. A Lucas-theorem argument shows that $f(pqr) = pq$ for suitable nearby primes $p < q < r$. The prime number theorem, combined with an elementary packing lemma, supplies unboundedly many such triples. This exact result is kernel-checked in Lean. This document makes no novelty or priority claim; the purpose is to expose the repository's proof in a compact, independently reviewable form.

1 Lucas's criterion

For a prime s , write $u \preceq_s v$ if every base- s digit of u is at most the corresponding digit of v . Lucas's theorem gives

$$s \nmid \binom{n}{k} \iff k \preceq_s n.$$

When $s \mid n$, the units digit of n is zero, and therefore

$$s \nmid \binom{n}{k} \iff s \mid k \text{ and } k/s \preceq_s n/s. \tag{1}$$

2 The nearby-prime structural lemma

Lemma 1. *Let*

$$q = p + a, \quad r = q + c = p + a + c, \quad b = a + c,$$

where $p < q < r$ are primes, $0 < a < c$, and $p > 4b^3$. Put $n = pqr$. For every $0 < k \leq n/2$, at most one of p, q, r fails to divide $\binom{n}{k}$.

Proof. The hypotheses make the following genuine base expansions, with the units digit written first:

$$qr = ab + (a + b)p + p^2, \tag{2}$$

$$pr = (q - ac) + (c - a - 1)q + q^2, \tag{3}$$

$$pq = bc + (p - c)r. \tag{4}$$

All displayed digits are nonnegative and smaller than their bases because $a, c < b$ and $p > 4b^3$.

Suppose first that both p and q are absent from $\binom{n}{k}$. By (1), $k = pqt$ for some positive integer t , and

$$qt \preceq_p qr, \quad pt \preceq_q pr. \tag{5}$$

The half-range gives $t \leq r/2$. Write $at = up + v$ with $0 \leq v < p$. Then

$$qt = v + (t + u)p,$$

and $t + u < p$: indeed $t < p$, $u < a$, and $t + u \leq (p + b)/2 + a - 1 < p$. The next-to-units comparison with (2) gives $t + u \leq a + b$. Hence $at < p$; the units comparison then gives $at \leq ab$, so $t \leq b$. Thus $at < q$ and

$$pt = (q - at) + (t - 1)q.$$

Comparison with (3) gives $t \geq c$ from the units digit and $t \leq c - a$ from the next digit, a contradiction.

Suppose next that q and r are both absent. Write $k = qrt$. Then

$$rt \preceq_q pr, \quad qt \preceq_r pq,$$

and $t \leq p/2$. Write $ct = uq + v$, $0 \leq v < q$. Since $t + u < q$, the first two base- q digits of rt are

$$rt = v + (t + u)q.$$

Comparison with (3) gives $t + u \leq c - a - 1$, and hence $t \leq c - a - 1$. In particular $ct < r$, so

$$qt = (r - ct) + (t - 1)r.$$

The units comparison with (4) would give $r - ct \leq bc$. But $ct + bc < 2b^2 < p < r$, a contradiction.

Finally suppose that p and r are both absent. Write $k = prt$. Then

$$rt \preceq_p qr, \quad pt \preceq_r pq,$$

and $t \leq q/2$. Write $bt = up + v$, $0 \leq v < p$. Again $t + u < p$, so the first two base- p digits of rt are $v, t + u$. Comparison with (2) gives first $t \leq a + b$, then $bt < p$, then $bt \leq ab$, and therefore $t \leq a$. Now

$$pt = (r - bt) + (t - 1)r.$$

Comparison with (4) would imply $r - bt \leq bc$, hence $r \leq ab + bc = b^2$, contrary to $r > p > 4b^3$. All three possible pairs lead to contradictions. $\square$

Lemma 2. *Under the hypotheses of Lemma 1,*

$$f(pqr) = pq \quad \text{and} \quad f(pqr)^2 > pqr.$$

Proof. The integer $n = pqr$ is squarefree. Lemma 1 says that every admissible gcd contains at least two of p, q, r , and is therefore at least pq . At $k = r$, (4) gives $pq \bmod r = bc \geq 1$, so $1 \preceq_r pq$ and criterion (1) shows that r is absent. The primes p and q cannot be absent because either absence would force the distinct prime to divide $k = r$. Thus the gcd at $k = r$ is exactly pq , proving $f(pqr) = pq$.

Finally

$$pq - r = p^2 + (a - 1)p - b > 0$$

under $p > 4b^3$. Hence $pq > r$ and $(pq)^2 > pqr$. $\square$

3 Producing infinitely many triples with the PNT

Lemma 3 (Asymmetric gaps in a short set). *If integers $x_0 < x_1 < \dots < x_{m-1}$ lie in an interval of length T and no $i < j < k$ satisfies*

$$x_k - x_j > x_j - x_i, \tag{6}$$

then $2^{m-2} < T$.

Proof. Put $S_j = x_{m-1} - x_j$. Failure of (6) for $(j, j+1, m-1)$ gives $S_{j+1} \leq x_{j+1} - x_j$, and therefore

$$S_j = (x_{j+1} - x_j) + S_{j+1} \geq 2S_{j+1}.$$

Since $S_{m-2} \geq 1$, iteration gives $x_{m-1} - x_0 = S_0 \geq 2^{m-2}$, while the interval hypothesis makes the span strictly less than T . $\square$

Theorem 4. *There are infinitely many composite integers n such that $f(n)^2 > n$.*

Proof. Let T tend through sufficiently large positive integers and put $N = 8T^3$. Partition $[N, 2N)$ into $8T^2$ half-open intervals of length T . The prime number theorem gives

$$\pi(2N) - \pi(N) \sim \frac{N}{\log N}.$$

Since

$$\frac{N/\log N}{8T^2(\log_2 T + 2)} \longrightarrow \infty,$$

some length- T interval contains more than $\lfloor \log_2 T \rfloor + 2$ primes. Lemma 3 then supplies primes $p < q < r$ in that interval with

$$r - q > q - p.$$

Put $a = q - p$, $c = r - q$, and $b = a + c = r - p$. Then $0 < a < c$, $b < T$, and

$$4b^3 < 4T^3 < 8T^3 \leq p.$$

Lemma 2 gives $f(pqr)^2 > pqr$. Moreover $pqr \geq (8T^3)^3$, so these products are unbounded as $T \rightarrow \infty$. An unbounded set of natural numbers is infinite. $\square$

4 Formal verification and claim boundary

The exact Lean declaration is `Erdos700PNT.erdos_700_ii` in `lean-part2/Solution.lean`, exposed through `lean-part2/Erdos700PNT.lean`. The three pair-omission cases in Lemma 1 are checked separately. The repository verifier rejects placeholders and reports only `propext`, `Classical.choice`, and `Quot.sound`, with no `sorryAx` dependency.

This establishes the proposition represented in the repository. It does not by itself establish novelty, priority, publication, or community acceptance. The original problem is from P. Erdős and G. Szekeres, “Some number theoretic problems on binomial coefficients,” *Australian Mathematical Society Gazette* 5 (1978), 97–99, archival PDF.